Class 12 Genome Informatics

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Section 4: Population Scale Analysis

> Q13. Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
expr <- read.table("https://bioboot.github.io/bggn213_W19/class-material/rs8067378_ENSG00000172057.6.tx
head(expr)
```

```
##    sample geno      exp
## 1 HG00367  A/G 28.96038
## 2 NA20768  A/G 20.24449
## 3 HG00361  A/A 31.32628
## 4 HG00135  A/A 34.11169
## 5 NA18870  G/G 18.25141
## 6 NA11993  A/A 32.89721
```

```
nrow(expr)
```

```
## [1] 462
```

```
table(expr$geno)
```

```
##
## A/A A/G G/G
## 108 233 121
```

```
tapply(expr$exp, expr$geno, median)
```

```
##      A/A      A/G      G/G
## 31.24847 25.06486 20.07363
```

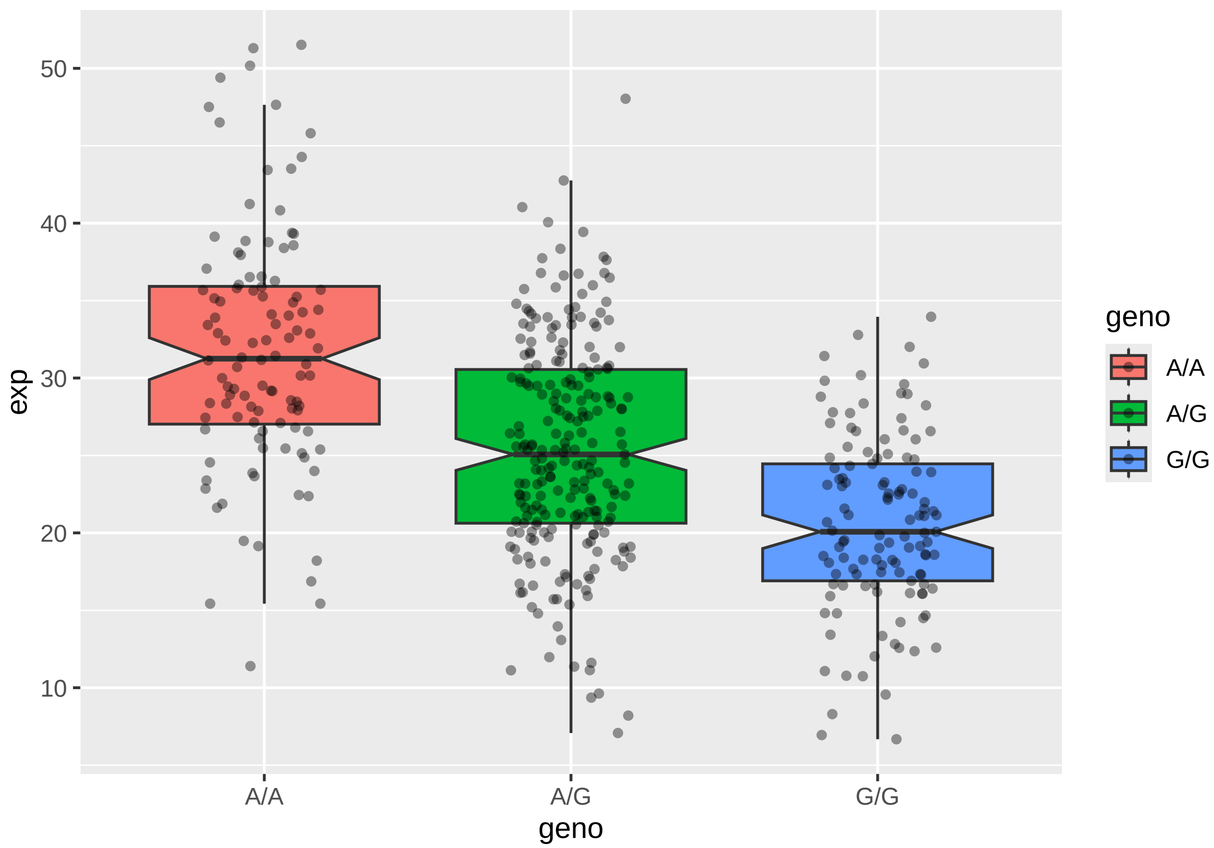
The sample sizes are 108 for A/A, 233 for A/G and 121 for G/G. The median expression levels are about 31.25 for A/A, 25.06 for A/G and 20.07 for G/G.

> Q14.Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)
```

Let's make a boxplot

```
ggplot(expr, aes(geno, exp, fill=geno)) +
  geom_boxplot(notch=TRUE, outlier.shape = NA) +
  geom_jitter(shape=16, position=position_jitter(0.2), alpha=0.4)
```



The ORMDL3 expression is highest in A/A, intermediate in A/G and lowest in G/G. Hence, this suggests the rs8067378 genotype is associated with ORMDL3 expression, with the G/G genotype showing lower expression.